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(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B. Tech. (CSE) (CC)

Semester: IV

SUBJECT CODE: BCSI402

OPERATING SYSTEMS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1 Answer the following questions (1 × 5 = 5 Marks)

(i) What is the primary role of the kernel in an operating system?
(ii) Name one common technique for managing deadlocks.
(iii) What does CPU scheduling aim to achieve?
(iv) Define the term "virtual memory."
(v) What is the purpose of a file system in an operating system?

Q. 2. Answer the following questions. **(1 × 5 = 5 Marks)**

(i) What is a process control block (PCB)?
(ii) Name one advantage of using threads over processes.
(iii) What is the function of a system call?
(iv) What is thrashing in the context of memory management?
(v) What is the difference between preemptive and non-preemptive scheduling?

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) Explain the Dining Philosopher Problem. What challenges does it present?

OR

(b) Describe the Banker's algorithm for deadlock avoidance.

(ii)(a) Draw & explain Process Control Block (PCB) with all its components. Also briefly explain why context switching is an overhead in the system?

OR

(b) Consider the following CPU process with arrival time (in milliseconds).

Process	Arrival Time	Burst time
P1	0	7
P2	3	3
P3	5	5
P4	6	2

If the pre-emptive shortest remaining time first scheduling algorithm is used to schedule the processes, then what is average waiting time?

Q.4.(i)(a) A system uses 3-page frames for storing process pages in main memory. It uses the Least Recently Used (LRU) page replacement policy. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below? 4, 7, 6, 1, 7, 6, 1, 2, 7, 2.

OR

(b) Explain the difference between internal and external fragmentation by giving suitable examples of each.

(ii) (a) Write short notes on the following:
(1) I/O Buffering. **(2)** Sequential File. **(03)** Indexed File.

OR

(b) Explain different issues in File System Implementation.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20Marks)

Q.5 (a) Consider the following reference string: 12342156212376321236
How many page faults will occur for?
1). FIFO
2). LRU Page Replacement Algorithm

OR

(b) What is partitioned memory allocation? Describe first-fit and best-fit strategies for disk space allocation, with their merits and demerits.

Q.6 (a) Draw the labeled process state transition diagram describing the various process states. Also, give the different reasons for process termination.

OR

(b) Explain layered structure of an operating system. Also explain advantages and disadvantages of the layered approach to system design.